

Java Basics Practical Solutions

Solution 1: `Book` Class

```
class Book {  
    private String title;  
    private String author;  
    private double price;  
  
    public Book(String title, String author, double price) {  
        this.title = title;  
        this.author = author;  
        this.price = price;  
    }  
  
    public String getTitle() { return title; }  
    public String getAuthor() { return author; }  
    public double getPrice() { return price; }  
}  
  
public class Main {  
    public static void main(String[] args) {  
        Book book1 = new Book("Java Programming", "James Gosling", 39.99);  
        Book book2 = new Book("Effective Java", "Joshua Bloch", 49.99);  
        System.out.println("Title: " + book1.getTitle() +  
            ", Author: " + book1.getAuthor() +  
            ", Price: $" + book1.getPrice());  
        System.out.println("Title: " + book2.getTitle() +  
            ", Author: " + book2.getAuthor() +  
            ", Price: $" + book2.getPrice());  
    }  
}
```

Solution 2: `Car` Class

```
class Car {  
    private String brand;  
    private int speed;  
  
    public Car(String brand, int speed) {  
        this.brand = brand;  
        this.speed = speed;  
    }  
  
    public void accelerate(int increase) {  
        speed += increase;  
    }  
  
    public void brake(int decrease) {  
        speed = Math.max(speed - decrease, 0);  
    }  
  
    public void displayInfo() {  
        System.out.println("Brand: " + brand + ", Speed: " + speed + " km/h");  
    }  
}  
  
public class Main {  
    public static void main(String[] args) {  
        Car myCar = new Car("Tesla", 50);  
        myCar.accelerate(30);  
        myCar.brake(60);  
        myCar.displayInfo();  
    }  
}
```

Solution 3: `Student` Class

```
class Student {  
    private String name;  
    private int age;  
    private double grade;  
  
    public Student(String name, int age, double grade) {  
        this.name = name;  
        this.age = age;  
        this.grade = grade;  
    }  
  
    public Student(String name, int age) {  
        this(name, age, 0.0);  
    }  
  
    public void displayInfo() {  
        System.out.println("Name: " + name + ", Age: " + age +  
            ", Grade: " + grade);  
    }  
}  
  
public class Main {  
    public static void main(String[] args) {  
        Student student1 = new Student("Alice", 20, 85.5);  
        Student student2 = new Student("Bob", 18);  
  
        student1.displayInfo();  
        student2.displayInfo();  
    }  
}
```

Java Static Practice Solutions

Solution 1: Employee Class

```
class Employee {  
    private String name;  
    private double salary;  
  
    private static String companyName = "Tech Corp"; // Shared across all  
employees  
  
    public Employee(String name, double salary) {  
        this.name = name;  
        this.salary = salary;  
    }  
  
    public static void setCompanyName(String newCompanyName) {  
        companyName = newCompanyName;  
    }  
  
    public void displayInfo() {  
        System.out.println("Employee: " + name + ", Salary: " + salary +  
            ", Company: " + companyName);  
    }  
  
    public static void main(String[] args) {  
        Employee e1 = new Employee("Alice", 50000);  
        Employee e2 = new Employee("Bob", 60000);  
        e1.displayInfo();  
        e2.displayInfo();  
  
        // Updating the company name  
        Employee.setCompanyName("NextGen Solutions");  
        e1.displayInfo();  
        e2.displayInfo();  
    }  
}
```

Solution 2: BankAccount Class

```
class BankAccount {  
  
    private static int totalAccounts = 0; // Shared across all accounts  
  
    private int accountNumber;  
  
    private double balance;  
  
    public BankAccount(double initialBalance) {  
  
        this.accountNumber = ++totalAccounts; // Assign a unique account number  
  
        this.balance = initialBalance;  
  
    }  
  
  
  
  
    public void deposit(double amount) {  
  
        if (amount > 0) {  
  
            balance += amount;  
  
        }  
  
    }  
  
  
  
  
    public void withdraw(double amount) {  
  
        if (amount > 0 && amount <= balance) {  
  
            balance -= amount;  
  
        } else {  
  
            System.out.println("Insufficient funds.");  
  
        }  
  
    }  
  
  
  
  
    public void displayInfo() {  
  
        System.out.println("Account Number: " + accountNumber +  
                            ", Balance: $" + balance);  
  
    }  
  
  
  
  
    public static void displayTotalAccounts() {  
  
        System.out.println("Total Accounts Created: " + totalAccounts);  
  
    }  
  
}
```

```
}
```

```
public static void main(String[] args) {  
    BankAccount acc1 = new BankAccount(1000);  
    BankAccount acc2 = new BankAccount(2000);  
    acc1.displayInfo();  
    acc2.displayInfo();  
  
    BankAccount.displayTotalAccounts();  
}  
}
```